MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

Triangular Combinatorics, Subitizing & Part-Whole Decompositions

SERIES 1 • CARD 02 / 12 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS

CARD 02 / 12 • DOMI-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

TRIANGULAR COMBINATORICS & PART-WHOLE PAIRS

Double-six dominoes form a triangular math system.

It contains all unordered pairs [a|b] from 0 to 6:

Total Tiles = T(7) = (7 × 8) / 2 = 28

Unique Tiles

Each tile is a 2-part additive partition of 0–12!

MODULAR 28.0 × 14.0 MM TILE DATUM

- Tile Size: 28.00 × 14.00 × 7.00 mm (4U×2U×1U)
- Tactile Pips: 3.20 mm ⌀ recessed high-contrast cups
- Center Divider: Recessed 1.2mm scoring groove

DOMINO SUBITIZING & SUM: [5|4] + [3|2] = 14



[5|4] = 9 [3|2] = 5

PART-WHOLE SUM

[5|4] pips = 9

[3|2] pips = 5

Total: 9 + 5 = 14

T(7) = 28 Unique Tiles Double-Six Combinatorics

2. KIT PARTS INVENTORY (33 PIECES)

COMPONENT	QTY	SPEC	COLOR
Double-Six Tiles	28x	28×14×7 mm	Stealth Black
Wildcard Blanks	2x	28×14×7 mm	Stealth Black
12-Frame Tray	1x	Dual 6-Bay	Matte Charcoal
Doubles Lid	1x	7-Slot Rail	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 30 Domino Tiles + 12-Frame Tray + Doubles Lid + Storage Bin = 33 Parts

HIGH-CONTRAST PIPS & TEXTURE

- Stealth Black Matte: Low glare, fingerprint resistant polymer.
- Brilliant White Pips: Standard dice quincunx configurations.
- Doubles Spine: Labeled slots [0|0] to [6|6] for organizing.
- High Durability: Scratch-resistant tactile surfaces.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Arrange the 28 standard tiles and 2 wildcards into 4 columns of 7 and 1 row of 2 on the floor. Missing tiles break the brick layout! Snap lid securely.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

The number of unordered pairs chosen with replacement from n elements is T(n) = n(n+1)/2.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: TRIANGULAR COMBINATORICS & PART-WHOLE SUMS

Step-by-Step Hands-On Tutorial: T(7) = 28 Tiles & [5|4] + [3|2] = 14

SERIES 1 • CARD 02 OF 12 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS

CARD 02 / 12 • DOMI-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/math-dominoes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS K.OA.A.3 (Decompose to 10)
- CCSS 1.OA.B.4 (Unknown Addend)
- TEKS 111.3.b.3.A (Model Additi...)

Grades K–5 Addition & Combinatorics

💡 GUIDING INQUIRY

If you flip a domino end-for-end, does the total pip sum change? Why or why not?

① STEP 1: LINE UP DOUBLES SPINE [0|0] TO [6|6]

Place the 7 double tiles in the lid's organizer spine:
[0|0]=0, [1|1]=2, [2|2]=4, [3|3]=6, [4|4]=8, [5|5]=10, [6|6]=12.
These 7 doubles form the diagonal spine of the 28-tile triangle!

② STEP 2: PAIR COMPLEMENTARY SUM-TO-6 TILES

Find tiles whose pip sum equals 6: [0|6], [1|5], [2|4], [3|3].
Place them in the 12-frame tray side by side.
Observe the inverse symmetry: as left side increases by 1,

③-④ STEPS 3 & 4: COMBINE TILES & REGROUP

3. COMBINE TWO TILES: Take tile [5|4] (sum 9) and [3|2] (sum 5). Seat in 12-frame tray.
4. CALCULATE SUM: 9 + 5 = 14! (1 full 10-frame + 4 ones).
Calculator Verification: Press Domino Pip Pattern Icon ([g][COUNT] & [g][a:b]) and run 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] → 14.

⑤ CORE PRINCIPLE: THE 28-TILE TRIANGULAR NUMBER SECRET

Why are there exactly 28 tiles? Count them by their highest pip: 7 tiles have a 0 ([0|0]..[0|6]),
6 new tiles have a 1 ([1|1]..[1|6]), 5 have a 2, down to 1 tile with a 6 ([6|6]).
Sum: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28! Dominoes are triangular numbers made tactile!

★ EXTENSION INVESTIGATION: FIND ALL SUM-TO-7 TILES

Can you find all 3 tiles that sum to 7: [1|6], [2|5], and [3|4]? Why is there no [0|7] tile in a double-six set? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL

stackcalc.io/parts

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5

Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING

Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1

Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4

Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

DOMI-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 33 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California